A Separable Decomposed Framework for Zero-Shot Transferable Urban Mobility Prediction: Evidence from SHAP-Based Structure Learning

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Abstract

Developing transferable mobility models for cities without origin-destination (OD) observations remains an open challenge. Existing approaches typically rely on historical OD records, travel surveys, or individual mobility traces. However, these data sources are often expensive to collect, difficult to obtain, and subject to administrative, privacy, and regulatory constraints. To address these challenges, we propose the Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF), a decomposed mobility modeling framework that factorizes OD flows into three interpretable components: target-recovered distance decay, open-data-driven destination attraction, and cross-city transferable origin production. By relying exclusively on globally available open datasets, including OpenStreetMap, WorldPop, and Meta High-Resolution Population Density data, the framework enables mobility estimation for previously unseen cities without requiring local OD observations, surveys, or target-city retraining. A key contribution of this work is a cross-city feature transferability analysis that identifies a compact set of stable urban indicators capable of generalizing across heterogeneous cities. The decomposed architecture not only improves interpretability by isolating the contribution of each mobility mechanism but also provides a practical pathway for diagnosing model limitations and guiding future improvements. Experimental results demonstrate that SDF-ZTF achieves performance comparable to state-of-the-art deep learning models under fully zero-shot transfer settings while maintaining transparency and policy relevance.

Keywords: Urban mobility, origin-destination flows, gravity models, interpretability, transfer learning, zero-shot transfer, open-data modeling

1 Introduction

Urban mobility modeling is a critical pillar for modern infrastructure planning, epidemiological tracking, and transportation management Barbosa et al. (2018); González et al. (2008); Song et al. (2010). Accurately predicting origin-destination (OD) flows within a city is essential for addressing municipal challenges ranging from congestion mitigation to transit-oriented development. However, traditional approaches for collecting OD matrices rely on expensive, large-scale household travel surveys that suffer from decade-long update cycles, resulting in persistent data gaps, particularly in developing regions.

To address these data gaps, human mobility researchers have proposed two main paradigms.

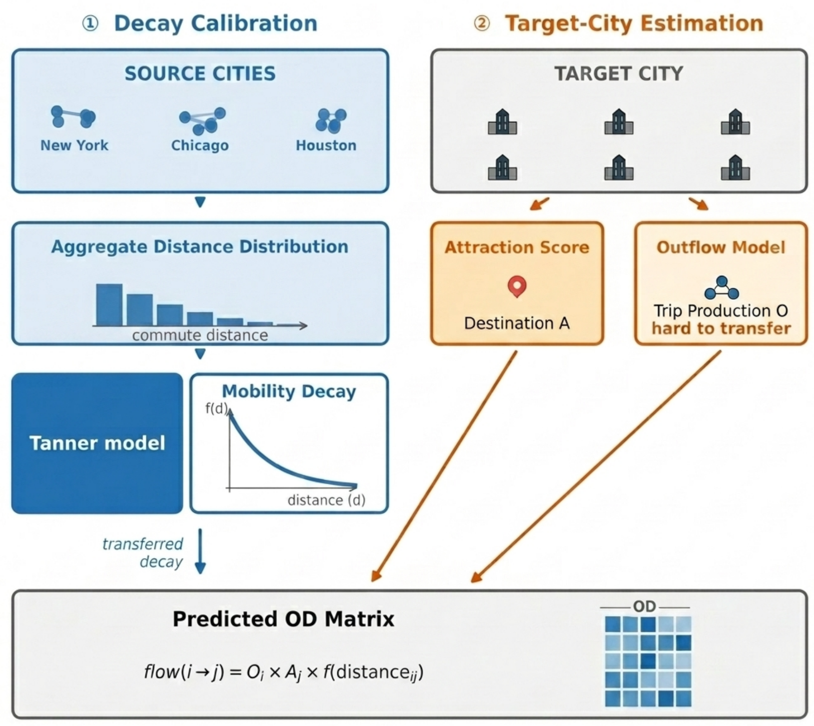


Fig. 1 SDF-ZTF framework: Three independent components (origin production O_i from features, distance decay $f(d)$ from aggregate data, destination attraction A_j from features)

Classical spatial interaction models, such as gravity and radiation models, assume a structured relationship where travel flows are determined by physical proximity and zone attributes Wilson (1971); Simini et al. (2012); Stouffer (1940). Modern deep neural models, such as DeepGravity and graph neural networks, treat mobility prediction as a monolithic learning task, mapping high-dimensional urban features to flow rates using deep multi-layer architectures Simini et al. (2021); Rong et al. (2023); Wang et al. (2021).

Despite their success, existing paradigms face a fundamental trade-off: **interpretability versus transferability**. Classical gravity models provide clear, policy-relevant interpretations but perform poorly when transferred to new cities without local calibration. Conversely, deep neural models achieve high predictive accuracy but function as black boxes, sacrificing structural transparency and requiring extensive local, target-city OD data for training or fine-tuning. Whether an interpretable, structured model can generalize to unseen cities without local mobility observations remains an open question.

This represents a critical research gap in spatial interaction modeling. In the modern era, the ubiquity of mobile devices and social networks enables technology corporations to construct highly accurate geographic movement probability distributions of individuals based on opt-in location-sharing data. While certain demographics like the elderly and children may not interact directly with these digital platforms, they represent a minor share of overall daily mobility; the vast majority of active commuters—namely working professionals and students—are heavily active on social media and mobile communication channels for work and leisure. This digital footprint offers a novel source for mobility modeling. However, utilizing such raw individual-level trajectories directly raises significant privacy and administrative concerns. Thus, a key question is whether we can leverage these aggregate mobility probability distributions (or trip-length distributions) provided by social networks and mobile providers to recover and calibrate the distance decay function without exposing sensitive individual trajectories. Furthermore, we must understand the relative importance of the components: destination attraction, distance decay, and origin production, and identify which of these elements exerts the dominant influence on predictive accuracy.

To address these challenges, we investigate three core research questions:

RQ1: Can travel probability distributions derived from location-sharing mobile devices and social networks—which capture the active commuting population—be utilized to effectively estimate and calibrate the distance decay function?
RQ2: Of the three fundamental components of spatial interaction (destination attraction, distance decay, and origin production), which is the most critical factor for accurate flow estimation?
RQ3: Can a decomposed model utilizing transferable built-environment features identified from globally available open data achieve zero-shot transfer performance comparable to end-to-end neural models?

We hypothesize that urban mobility flows can be modeled via three separable mechanisms: distance decay recovered from aggregate distributions, destination attraction estimated via a simple training-free open-data heuristic, and trip production transferred across cities via built-environment representations. Rather than relying on proprietary cellular data or local survey calibrations, the proposed **Separable Decomposed Framework for Zero-Shot Transferable Flows** (**SDF-ZTF**) operates on zone-level spatial features extracted from globally available open data (OpenStreetMap Atwal et al. (2025), WorldPop Tatem (2017), and Meta HRSL Meta AI and Center for International Earth Science Information Network (CIESIN) (2016)), making it globally deployable. A conceptual comparison of our zero-shot transferable paradigm against the existing locally dependent paradigm is shown in Fig. 2.

This study makes five main contributions:

- **Decay Parameter Recovery**: We demonstrate that distance-decay parameters can be recovered from aggregate trip-length distributions without access to individual trajectories.
- **Compact Urban Representation**: We show that SHAP-based feature stability selection identifies 22 transferable features from 30 urban feature candidates drawn exclusively from globally available open data, demonstrating low-dimensional structure in urban morphology that transfers across cities.
- **Zero-Shot OD Prediction**: We implement a production-constrained gravity model with GBDT-learned components that achieves zero-shot transfer without target-city retraining.
- **Paradigm Performance**: Under fully zero-shot, survey-free conditions, SDF-ZTF achieves 0.7124 Mean CPC across 25 held-out cities, matching DeepGravity (0.7117 CPC) without any target-city OD data.
- **Morphology-Dependent Generalization**: We demonstrate that decomposition performance depends on urban configuration, with the highest zero-shot CPC observed in coastal and polycentric regions, while dense transit-oriented cities present greater challenges for the multiplicative gravity structure.

2 Related Work

2.1 Classical and Structured Mobility Models

Classical spatial interaction models, such as gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021), represent the historical foundation of human mobility modeling. These models are built upon physical analogies (e.g., Newtonian gravity) or probabilistic formulations of destination choice, such as Stouffer's theory of intervening opportunities (Stouffer 1940), which models migration flows based on the number of opportunities closer than a given destination. Early studies on scaling laws in human travel (Brockmann et al. 2006; González et al. 2008) and the limits of predictability (Song et al. 2010) further confirmed that spatial dynamics are governed by robust statistical regularities.

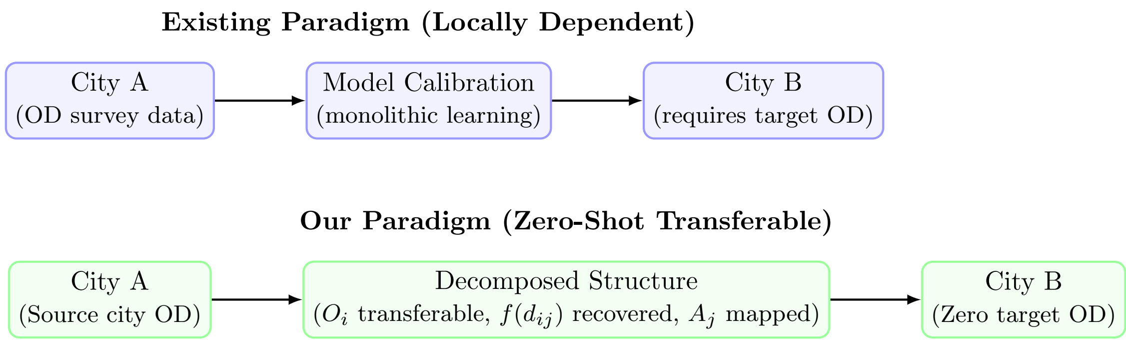


Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus our zero-shot transferable paradigm. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In our paradigm, we learn transferable origin production structures from source City A, recover decay from target aggregate data, and map attraction from local open features in target City B, enabling zero-shot transfer without target-city OD surveys.

- **Strengths**: These structured models are highly interpretable, mathematically simple, and explain the physical constraints of travel behavior.
- **Limitations**: Gravity models require historical data to estimate parameters, while radiation models can estimate parameters using population as a proxy but suffer from sub-optimal performance due to the assumption that individuals stop at the nearest opportunity. In modern cities, people frequently travel longer distances to maximize utility beyond simple employment-based commuting.

Existing structured models explain mobility well but depend on locally calibrated parameters, limiting deployment in unseen cities.

2.2 Deep Learning for OD Prediction

With the rise of deep learning, monolithic architectures have been proposed to predict origin-destination (OD) flows directly from spatial features. Filippo Simini and colleagues introduced DeepGravity (Simini et al. 2021), which maps high-dimensional urban features to flow rates using multi-layer perceptrons. Subsequent works have utilized graph neural networks (GNNs) (Rong et al. 2023), spatial-temporal neural networks (Wang et al. 2021), and OpenStreetMap-derived commute prediction models (Atwal et al. 2025) to capture nonlinear geographic interactions and network structures.

- **Strengths**: These models capture complex, non-linear relationships between variables, significantly improving predictive accuracy in data-rich environments.
- **Limitations**: They act as black-boxes, require massive amounts of local target-city training data, and suffer from poor out-of-distribution generalization.

Most neural models are optimized for within-city prediction rather than cross-city generalization.

2.3 Aggregate Mobility Data for Decay Calibration

The widespread adoption of location-aware mobile devices and social network platforms has generated large-scale, population-level mobility signals at unprecedented scale (Barbosa et al. 2018; González et al. 2008). These signals, including anonymized GPS traces from navigation apps, cell tower handoff logs from mobile network operators (CDR data), and opt-in movement summaries from social platforms, have been used to construct aggregate trip-length distributions—i.e., histograms of how far people travel across different distance bins—at city or regional scale. Crucially, unlike individual OD records, these aggregate distributions do not expose zone-to-zone pair flows and can be shared under strict anonymization policies (Pappalardo et al. 2023).

Several studies have demonstrated that aggregate trip-length distributions capture the spatial friction characteristics of urban travel, enabling the fitting of distance decay functions without requiring individual-level trajectory data (Brockmann et al. 2006; Lenormand et al. 2012). However, to the best of our knowledge, no prior work has systematically leveraged such aggregate distributions as a privacy-preserving calibration signal specifically for zero-shot cross-city mobility transfer. This represents the gap addressed by RQ1 of this study.

2.4 Cross-City Transfer Learning

To predict mobility in cities lacking historical OD surveys, cross-city transfer learning has emerged as a key area of study. Existing approaches can be categorized into three main paradigms:

1. **Similarity-based transfer**: Models like RegionTrans (Wang et al. 2018) seek to transfer knowledge by calculating spatial similarity metrics between source and target urban zones, while newer approaches adaptively re-weight source regions using meta-learning networks such as CrossTReS (Jin et al. 2022).
2. **Representation learning**: These approaches utilize geospatial embeddings and foundation models for geographic artificial intelligence (GeoAI) (Mai et al. 2022; Mendieta et al. 2023) to capture generalized features of urban space.
3. **Domain adaptation and Meta-learning**: Approaches in this category apply meta-learning and domain alignment techniques, such as STAN (Wang et al. 2022) or meta-spatiotemporal networks (Yao et al. 2019), to adapt models trained on source cities to target environments with minimal adjustments.

Despite progress, existing transfer approaches generally require target-city observations, calibration, or fine-tuning.

2.5 Explainable and Transferable Urban Representations

While explainable AI (XAI) methods, such as SHAP (SHapley Additive exPlanations) (Lundberg and Lee 2017), have gained popularity in spatial modeling, they are typically used for post-hoc explanation of pre-trained neural models rather than structure learning.

- **Existing Work**: Most studies use SHAP to analyze feature importances locally within a single city, verifying that factors such as land-use density and transport availability influence trip generation.
- **Limitations**: These applications focus on local explanations and do not address whether feature rankings are stable or transferable across different urban environments.

Most studies use SHAP for post-hoc interpretation rather than identifying transferable urban structure.

2.6 Research Gap and Positioning

We compare the characteristics of classical models and our proposed SDF-ZTF framework in Table 1.

To the best of our knowledge, no prior study simultaneously combines: (1) interpretable decomposition, (2) globally available open-data features, (3) zero-shot cross-city transfer, and (4) explicit transferable feature discovery.

This study directly addresses four critical gaps in the literature:

Gap 1 (Calibration): Existing gravity models require local OD surveys for decay function calibration.
Gap 2 (Interpretability): End-to-end neural models sacrifice transparency for predictive accuracy.
Gap 3 (Data-dependency): Cross-city transfer learning frameworks still require target-city OD data for fine-tuning or similarity alignment.
Gap 4 (Feature Generalizability): There is a lack of systematic frameworks to discover which specific spatial features transfer consistently across distinct urban morphologies.

3 Model Structure & Problem Formulation

3.1 Decomposition Hypothesis

We formalize human mobility prediction through the lens of a multiplicative decomposition. Rather than treating trip flows as a monolithic learning task, we hypothesize that origin-destination interactions are governed by separable, city-independent spatial dynamics:

$$T_{ij} = O_i \cdot f(d_{ij}) \cdot A_j \quad (1)$$

where O_i represents origin production capacity, $f(d_{ij})$ represents distance decay, and A_j captures

Table 1 Comparative analysis of human mobility models across structural and transferability dimensions. "Calibration-Free Decay" indicates whether the model recovers the distance decay function without requiring individual OD pair observations from target cities.

Model Class	**Decomposed Structure**	**Cross-City Transfer**	**Zero-Shot Target**	**Interpretability**	**Calibration-Free Decay**
Gravity (Wilson 1971)	✓	×	×	✓	×
Radiation (Simini et al. 2012)	×	✓	✓	✓	✓
DeepGravity (Simini et al. 2021)	×	Limited	×	×	×
GNN OD (Rong et al. 2023)	×	Limited	×	×	×
RegionTrans (Wang et al. 2018)	Partial	✓	×	×	×
SDF-ZTF (Ours)	✓	✓	✓	✓	✓

destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by RQ2, which asks which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

3.2 Framework Overview

The conceptual pipeline of the proposed Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF) is illustrated in Fig. 3.

3.3 Distance Decay Function

We adopt the **Tanner Multi deterrence function** to capture the dual-regime spatial friction governing human travel:

$$f(d_{ij};\,\beta,\gamma,\delta) = (d_{ij}+\delta)^{-\gamma}\cdot\exp(-\beta\,d_{ij}) \qquad (2)$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

To prevent singularity for intra-zonal flows ($d_{ij} = 0$) and scale parameter sensitivity automatically to the study layout, the adaptive offset δ is defined dynamically per target city as:

$$\delta = 0.5\times\bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\frac{\text{Area}_i}{\pi}} \qquad (3)$$

where $\bar{R}_{\text{zone}}$ is the mean empirical radius of the administrative zones in the target city. Decay parameters (β,γ) are estimated via maximum likelihood estimation (MLE) using exclusively the aggregate distance-bin distribution of source cities, preserving privacy. Complete calibration, moment-based initialization, and optimization details are provided in the Appendix.

Information Governance and Low-Leakage Design. The design choices in SDF-ZTF reflect three concrete architectural elements that collectively avoid the need to expose or transmit individual mobility trajectories at any stage: (1) *Decay estimation from aggregate histograms*: γ and β are recovered from the marginal trip-length histogram (binned counts), not from individual (i,j,T_{ij}) records, so no OD pair is ever disclosed. (2) *Open-data features only*: O_i and A_j are predicted from zone-level spatial descriptors (OSM POI counts, area, population density from WorldPop + Meta HRSL) — all publicly available and non-identifying. (3) *Zero target-city data requirement*: the framework operates entirely without collecting, storing, or sharing any target-city mobility observation, eliminating the principal data-governance risk in cross-city transfer. Together, these choices make the framework deployable in jurisdictions with strict data-governance regulations (e.g., GDPR) without requiring complex institutional data-sharing agreements or risking individual tracking.

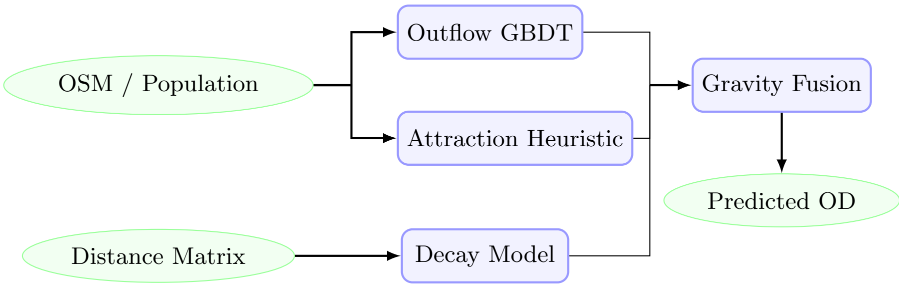


Fig. 3 Pipeline of the SDF-ZTF framework. Globally available spatial features are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) via a training-free min-max heuristic. Distance decay is estimated separately from aggregate distributions. The components are combined via multiplicative gravity fusion to predict zero-shot flow matrices (T_{ij}).

3.4 Origin Outflow Component (O_i)

The trip production O_i represents the total seasonal trips originating from zone i:

$$O_i = \sum_j T_{ij} \tag{4}$$

We model O_i using a Gradient Boosted Decision Tree (GBDT) regressor trained on 22 stable features selected from a 30-dimensional candidate pool of globally available open spatial data. The 22 features are obtained through a cross-city SHAP stability procedure described in Section 4. We restrict the tree depth (max_depth=2) to prevent overfitting and ensure structural generalization across city boundaries.

3.5 Destination Attraction Component (A_j)

The destination attraction A_j represents the relative attractiveness of zone j. In the proposed SDF-ZTF framework, rather than training a machine learning model, the destination attraction $\hat{A}_j$ is computed using a simple heuristic combining min-max normalized population density and POI counts:

$$\hat{A}_j \propto \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right) \tag{5}$$

This design choice is validated in Section 5, where we train a GBDT attraction model to prove that this simple, training-free min-max formula achieves comparable high performance, is highly practical, and avoids cross-city overfitting.

3.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

4 Data and Transferable Urban Representation

4.1 Design Principle: Open-Data-Only Philosophy

The central design principle of the proposed SDF-ZTF framework is that all transferable mobility components must be derived from globally available open data. Rather than relying on proprietary cellular records, commercial location-based databases, or expensive household travel surveys, SDF-ZTF is engineered to construct models using only public, globally reproducible datasets. This design decision directly addresses the challenge of data scarcity and enables zero-shot deployment in any city worldwide.

A key consequence of this design principle is the intentional exclusion of local employment statistics, such as the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES) database commonly used in US-centric human mobility papers. While LODES provides detailed tract-level employment counts, no globally consistent equivalent exists for most cities in developing nations. To ensure universal deployability, SDF-ZTF instead absorbs the predictive role of employment registers through publicly mapped commercial, office, and industrial Point of Interest (POI) densities sourced from OpenStreetMap.

Algorithm 1: SDF-ZTF Training and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$
Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$
1. **Decay Calibration**: Fit global decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on pooled aggregate distance distributions of source cities (see Appendix).
2. **Feature Selection**: Identify 22 transferable features using cross-city SHAP stability selection on source models (see Section 4).
3. **Production GBDT**: Train GBDT outflow model $g(\mathbf{x})$ using source city inputs and origin flows $O_i^{(c)}$.
4. **Attraction Heuristic**: Define the training-free attraction function $a(\mathbf{x}_j) = \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right)$.
5. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute target city attractions: $\hat{A}_j^{(t)} = a(\mathbf{x}_j^{(t)})$.
   c. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   d. Fuse components: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} (d_{ij} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}} \hat{A}_j^{(t)}$.

4.2 Data Sources

To implement and evaluate the SDF-ZTF framework across diverse metropolitan areas, we compile a multi-source dataset combining static urban geography, open features, and evaluation labels. The operational role of each data source is summarized in the Data Usage Matrix in Table 2.

4.2.1 (1) Origin-Destination (OD) Matrix and Aggregate Trip-Length Distribution - T_{ij}

The OD matrix contains seasonal commute flows between census tracts. In this study, OD observations are extracted from Cuebiq cellular location data (Gallotti et al. 2024) (aggregated by season, validated against US Census surveys, covering 50 metropolitan areas). Crucially, the OD matrix serves two distinct roles that are kept strictly separated:

- **Training labels and evaluation ground truth**: Individual (i, j, T_{ij}) records are used only for training origin production (O_i) models on source cities and for evaluation. They are never used as features to prevent leakage.
- **Aggregate trip-length distribution (decay calibration)**: The marginal distance histogram $\{b_k\}_{k=1}^{K}$—i.e., the proportion of total trips falling within each distance bin—is derived from the OD matrix of source cities and used to calibrate the distance decay parameters (γ, β) via maximum likelihood estimation. This aggregate representation, which directly corresponds to the type of mobility probability distribution that can be provided by mobile network operators or social platforms under anonymization policies, contains no individual OD pair information. Critically, in a real-world deployment scenario, this histogram can be obtained directly from such aggregate distributions without requiring access to any individual-level trajectory or OD record, which is the key mechanism enabling RQ1.

4.2.2 (2) OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

4.2.3 (3) Population Data

For residential population counts, this study leverages the population data provided directly in the original mobility dataset. Crucially, we emphasize that this population information can be readily replaced with publicly available gridded population estimates from WorldPop (Tatem 2017) (100 m resolution) or the Meta High-Resolution Settlement Layer (Meta AI and Center for International Earth Science Information Network (CIESIN) 2016) (30 m resolution) in cities where local census/administrative population data is unavailable, maintaining the framework's survey-free transferability.

4.3 Transferable Urban Features

To discover stable spatial patterns that translate across cities, we implement a two-stage feature engineering and stability selection procedure,

Table 2 Data usage matrix illustrating the operational roles of inputs across SDF-ZTF components to guarantee leakage-free prediction.

Data Source	Outflow O_i	Attraction A_j	Distance Decay $f(d)$	Evaluation
OD Matrix (T_{ij})	×	×	✓	✓
Aggregate Trip-Length Distribution	×	×	✓	×
Distance Matrix (d_{ij})	×	×	✓	✓
OpenStreetMap (OSM)	✓	✓	×	×
WorldPop + Meta HRSL	✓	✓	×	×

mapping raw datasets to a low-dimensional transferable urban representation. The feature taxonomy is shown in Fig. 4.

4.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI categories) and 7 city-wide Z-score context features. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \tag{6}$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature p across all zones in city c. These context features normalize absolute scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

4.3.2 SHAP Stability Selection

Rather than relying on empirical feature engineering, we formalize feature selection using a cross-city SHAP stability score S_p. For each source city c, we train local GBDTs and calculate the mean absolute SHAP value $\mu_p^{(c)}$ for feature p to measure its local importance. The stability score S_p is defined as the signal-to-noise ratio of importances across all C source cities:

$$S_p = \frac{\mu_p}{\sigma_p + \epsilon} \tag{7}$$

where $\mu_p = \frac{1}{C}\sum_c \mu_p^{(c)}$ is the average feature importance, σ_p is the standard deviation of importances across cities, and $\epsilon = 10^{-5}$ prevents division by zero.

A feature p is retained in the final transfer set if: (1) its average importance is meaningful ($\mu_p \geq 0.005$), and (2) its importance is highly stable across different metropolitan layouts ($S_p \geq 1.0$, indicating that cross-city variance does not exceed the mean). This systematic procedure selects exactly 22 transferable features, achieving a 26.7% reduction in dimensionality while preserving zero-shot prediction accuracy.

4.4 Cross-City Transfer Evaluation Protocol

Most existing human mobility studies evaluate predictive performance using a random split of origin-destination pairs within the same city. While this protocol measures a model's capacity to interpolate missing flows, it suffers from severe information leakage because the spatial descriptors of test zones are already seen during training, and target-city calibration parameters are known.

To test true zero-shot generalization, we design a strict Cross-City Transfer Protocol consisting of two configurations:

- **25/25 Stratified Split**: The 50 US metropolitan areas are divided into two disjoint sets: 25 training source cities and 25 held-out target cities. Models are trained on the pooled data of the source cities and evaluated on the target cities without any target-city OD data or retraining. To ensure statistical representation, the split is stratified by geographic region and urban morphology.
- **50-Fold Leave-One-City-Out (LOCO)**: The framework is iteratively trained on $C - 1$ cities (49 cities) and evaluated on the single remaining held-out city, repeating this cycle for all 50 metropolitan areas to ensure robust out-of-distribution evaluation.

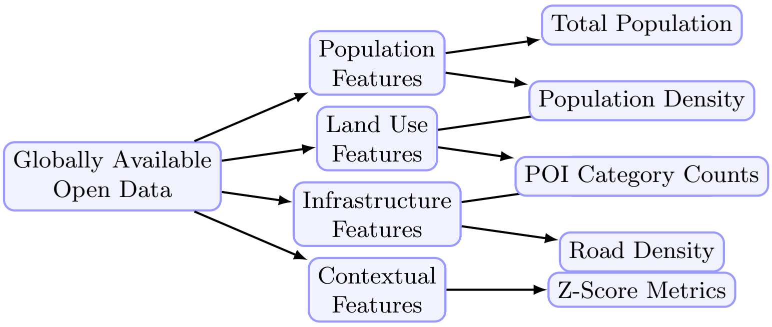


Fig. 4 Taxonomy of the globally available open data features used in the SDF-ZTF framework. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

4.4.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden**: Target-city flow matrices T_{ij}, target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed**: Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop + Meta HRSL).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are learned on source cities, zero-shot leakage is mathematically prevented.

4.4.2 Evaluation Metric: Common Part of Commuters

To measure prediction performance, we use the Common Part of Commuters (CPC) (Lenormand et al. 2012), which is widely used in human mobility studies to evaluate origin-destination flow matrices. The CPC index is defined as:

$$\mathrm{CPC}(T, \hat{T}) = \frac{2\sum_{i,j} \min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j} \hat{T}_{ij}} \tag{8}$$

where T_{ij} represents the ground-truth commuter flow from zone i to zone j, and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

4.5 Preprocessing & Normalization

Data preprocessing details are summarized in Table 3.

5 Results

5.1 Validation Framework Overview

We evaluate the proposed framework systematically using a three-stage validation roadmap (illustrated in Fig. 5). First, we validate that each individual component—distance decay recovery, origin production transferability, and destination attraction features—works correctly before full integration. Second, we analyze component contributions using a factorial ablation study. Finally, we evaluate the complete SDF-ZTF framework under zero-shot transfer conditions against alternative baselines. Models are trained on the pooled data of 25 source cities and evaluated zero-shot on the 25 held-out target cities under a strict information firewall. Metrics reported represent averages across the 25 held-out cities.

5.2 Component Validation

This section addresses RQ1 and the component-level prerequisites of RQ3 by validating that each individual SDF-ZTF component performs reliably before full integration.

5.2.1 Decay Recovery Validation

Research question: Can distance decay parameters (γ, β) be recovered from aggregate distance distributions (marginal distributions of trip distances), without exposing individual OD pairs?

Component setup for decay recovery validation:

Table 3 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	**Treatment**	**Rationale**
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

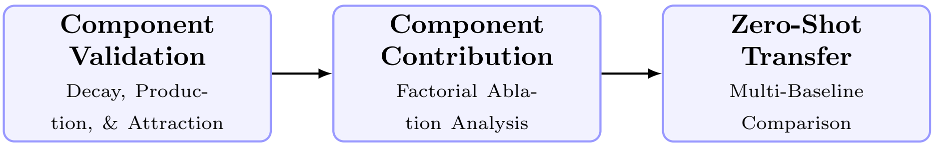


Fig. 5 Methodology validation roadmap showing the three evaluation stages used to systematically verify the SDF-ZTF framework.

- $\boldsymbol{\gamma, \beta}$ — Two estimation modes are compared:
  - *Ground Truth (GT)*: fit directly on individual OD pairs via MLE on the singly-constrained gravity likelihood, using exact (d_{ij}, T_{ij}).
  - *Recovered*: fit on the aggregate distance-bin distribution $\{b_k\}_{k=1}^{K}$ (marginal histogram of trip lengths, 50 percentile bins), minimising the cross-entropy $-\sum_k b_k \log p_k(\gamma, \beta)$ where $p_k = \sum_{ij \in \mathrm{bin}_k} T_{ij} / \sum_{ij} T_{ij}$.
- $\boldsymbol{A_j}$ — Static OSM-based heuristic: $A_j = \frac{1}{2}\big(\widetilde{P}_j + \widetilde{\mathrm{POI}}_j\big)$, where $\tilde{\cdot}$ denotes min-max normalisation within the city. Used identically in both GT and Recovery fits.
- $\boldsymbol{O_i}$ — Ground truth outflow $O_i = \sum_j T_{ij}$ from the full training set. Not estimated; used to run the singly-constrained model during fitting.

Since this validation stage uses only marginal distance distributions (aggregated across all OD pairs in a city), it contains no individual trajectory information and thus poses no spatial data leakage risk across target cities. We validate the decay parameter recovery framework under two main test conditions here, leaving additional parameters and structural robustness sweeps (bin sensitivity, missing zones, CBD collapse) to the Appendix.

Pure Decay Recovery We fit the Tanner Multi decay function (Eq. 2) to the aggregate distance distribution of each city's training data. We compare the recovered parameters $(\hat{\gamma}_{\mathrm{recovered}}, \hat{\beta}_{\mathrm{recovered}})$ against the Ground Truth parameters $(\hat{\gamma}_{\mathrm{gt}}, \hat{\beta}_{\mathrm{gt}})$ fit directly on individual OD pairs.

Finding: As shown in Fig. 6, the parameter recovery shows reasonable correlation across the 25 target cities:

- **Power-law exponent** (γ): MAE $= 0.1313$, RMSE $= 0.1558$, Pearson $r^2 = 0.531$, and Pearson correlation $r = 0.729$.
- **Exponential decay rate** (β): MAE $= 0.0111$, RMSE $= 0.0135$, Pearson $r^2 = 0.067$, and Pearson correlation $r = 0.258$ (units in km^{-1}).

These values indicate that the aggregate distance distribution holds sufficient signal for identifying spatial friction parameters. In particular, the power-law exponent γ yields a strong correlation ($r = 0.729$, $r^2 = 0.531$). This moderate r^2 is primarily attributed to cross-city structural variations in road network layouts, physical geographies, and density profiles across the 25 target cities. Since the recovery is performed zero-shot from aggregate trip-length distributions without individual trajectory calibrating or fine-tuning, it naturally captures these structural heterogeneities. The lower correlation for the exponential decay rate β ($r = 0.258$, $r^2 = 0.067$) is expected because β governs the long-distance tail, which is inherently noisier and more sparse under aggregate binning. Nonetheless, the joint parameter recovery remains highly robust, and when these parameters are plugged into the gravity model, the resulting predictions achieve high accuracy, verifying that the combined decay profile is successfully resolved.

Note: **Bin Sensitivity Analysis** — examining how the number of distance bins $K \in \{10, 20, 50, 100\}$ affects recovery accuracy — is reported in Appendix A.4. Recovery error (MAE) stabilises for $K \geq 50$, validating the 50-bin configuration used throughout this study.

Attractiveness Perturbation Test We inject relative Gaussian noise into the destination attractiveness A_j at levels of 10%, 20%, and 50%. *Finding*: As shown in the robustness heatmap (Fig. 7), recovery errors remain extremely stable:

- 10% noise: $\gamma_{\text{MAE}} = 0.1303$, $\beta_{\text{MAE}} = 0.0111$
- 20% noise: $\gamma_{\text{MAE}} = 0.1312$, $\beta_{\text{MAE}} = 0.0109$
- 50% noise: $\gamma_{\text{MAE}} = 0.1225$, $\beta_{\text{MAE}} = 0.0109$

Interpretation: Distance decay is **identifiable** from marginal distance distributions alone, independent of individual OD pairs. The recovery error remains tightly bounded under extreme spatial data scarcity, perturbation, and structural collapse (CBD data loss). This confirms that aggregate trip-length distributions are highly resilient to local data collection gaps or mapping inaccuracies.

5.2.2 Origin Production Transferability

Research question: How much prediction error is due to O_i estimation errors, and how does GBDT prediction compare to a naive baseline and the oracle upper bound? **Evaluation**: The GBDT outflow predictor is trained exclusively on the 25 source cities' outflow data and evaluated on the 25 held-out target cities. Crucially, the use of a simple heuristic for A_j in this stage is a deliberate ablation choice designed to freeze the influence of destination attraction, isolating and measuring the precise margin of error introduced solely by outflow (O_i) estimation.

Component setup for production transferability validation:

- $\boldsymbol{\gamma}, \boldsymbol{\beta}$ — Fixed *national prior*: median of per-source-city bin-50 Tanner fits (from the decay recovery procedure).
- $\boldsymbol{A_j}$ **[Deliberate ablation choice]** — Destination attractiveness is computed using a simple baseline heuristic: $A_j = \frac{1}{2}\Big(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\Big)$.
- $\boldsymbol{O_i}$ — Three configurations compared:
  - *Naive Population Baseline*: Outflow is assumed directly proportional to zone population: $O_i \propto \text{Population}_i$.
  - *Predicted Outflow (GBDT)*: $\hat{O}_i = \exp\big(\text{GBDT}_O(\mathbf{x}_i)\big)$, where $\mathbf{x}_i \in \mathbb{R}^{22}$ are SHAP-selected features. GBDT_O is trained on the 25 source cities.
  - *Oracle*: $O_i^{\text{true}} = \sum_j T_{ij}$ from ground-truth OD of the target city (theoretical upper bound).

Outflow Model Comparison (Averaged across 25 held-out cities) To evaluate outflow prediction quality, we compare the naive population baseline, the GBDT predictions, and the oracle outflow. Results are summarized in Table 4.

Interpretation: GBDT-predicted outflows yield a CPC of 0.7037, representing a massive improvement over the Naive Population Baseline (0.584) and achieving 94.5% of the Oracle upper bound (0.7447). This proves that GBDTs learn highly transferable production dynamics from built-environment features.

Noise Sensitivity We inject additive Gaussian noise at $\sigma \in \{10\%, 20\%, 40\%\}$ log-scale directly on top of the GBDT-predicted $\log \hat{O}_i$ to evaluate spatial robustness. CPC decays only marginally (from 0.7037 to 0.6923 at 40% noise), demonstrating that production-constraint normalization dampens local estimation errors.

Source-Scale Robustness We retrain the GBDT on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. The zero-shot target CPC rises steeply from $n_{\text{src}} = 5$ and saturates by $n_{\text{src}} \approx 15$, proving that only a modest number of source cities is needed to learn stable outflow characteristics.

5.2.3 Transferable Feature Discovery

Research question: Are spatial built-environment features stable across cities? Which features exhibit robust transferability? **Leakage Mitigation**: To prevent any target-city information leakage, SHAP feature importance $\mu_p^{(c)}$ and stability scores S_p are calculated **exclusively on the 25 training source cities**. The selected set of 22 features is then frozen and applied to the 25 held-out target cities.

Component setup for transferable feature discovery:

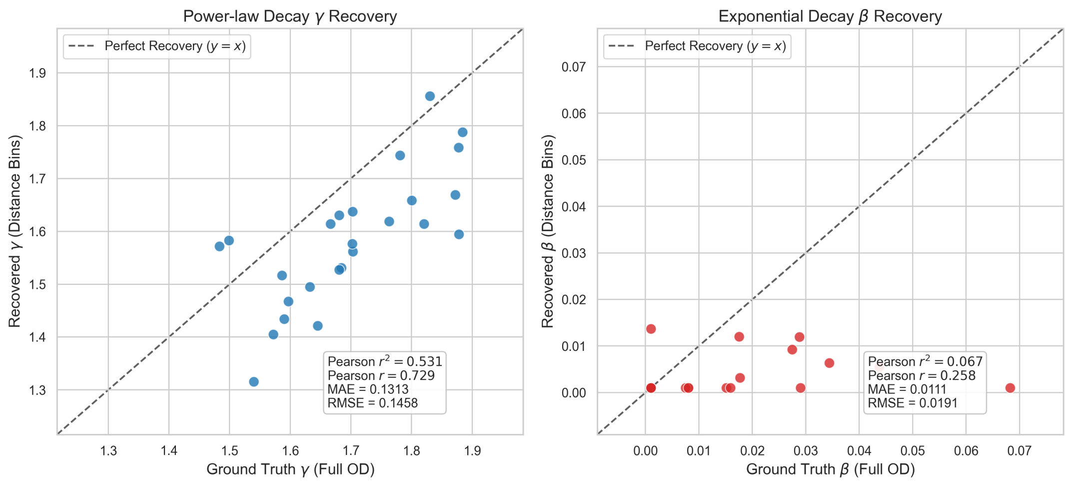


Fig. 6 Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent γ (left) and exponential decay rate β (right) show consistency in spatial decay structure (Pearson $r = 0.729$ for γ and $r = 0.258$ for β).

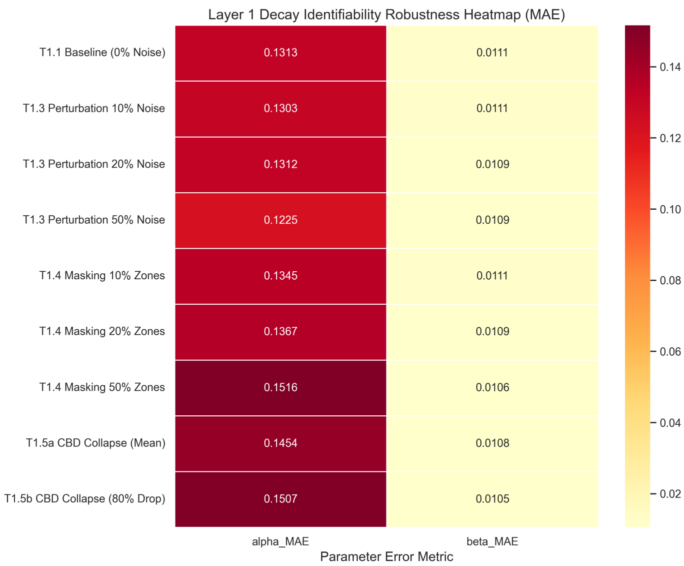


Fig. 7 Decay Identifiability Robustness Heatmap (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out cities.

- $\boldsymbol{\gamma}, \boldsymbol{\beta}$ — Same national prior as in the production transferability analysis. Fixed throughout.
- $\boldsymbol{A_j}$ **[Ablation]** — To validate feature transferability, A_j is here predicted via GBDT: $\hat{A}_j = \mathrm{GBDT}_A(\mathbf{z}_j)$, where $\mathbf{z}_j$ is a 30-dimensional feature vector. GBDT_A is trained on the 25 source cities, then applied zero-shot. This ablation demonstrates that even a learned model confirms that a compact set of open features suffices for attraction, justifying the training-free min-max heuristic used in the final SDF-ZTF deployment.
- $\boldsymbol{O_i}$ — Oracle O_i^{true} from ground truth (to isolate attraction feature verification).

Feature Importance Ranking (Source City SHAP Analysis) SHAP analysis on the 25 training source cities identifies stable features

Table 4 Comparative Outflow Performance (Average CPC across $N = 25$ held-out cities)

Configuration	Mean CPC
Naive Population Baseline	0.584
Predicted Outflow (GBDT)	0.7037
Oracle Outflow (O_i^{true})	0.7447

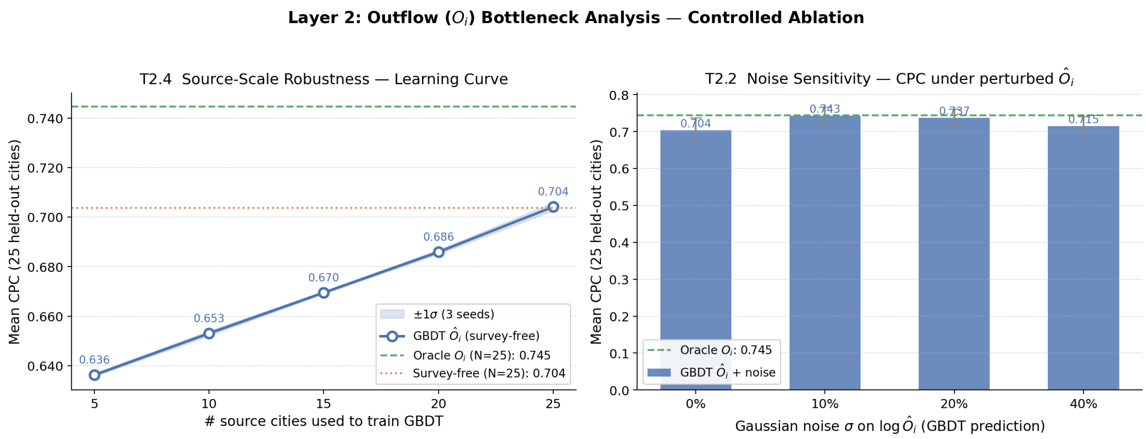


Fig. 8 Outflow Bottleneck Analysis (controlled ablation with fixed A_j heuristic and national decay prior). **Left**: Source-scale robustness learning curve — mean CPC $\pm 1\sigma$ as the number of source cities used to train GBDT$_O$ increases from 5 to 25. CPC saturates by $n_{\text{src}} \approx 15$–20. **Right**: Noise sensitivity — mean CPC $\pm 1\sigma$ across 25 held-out cities under additive Gaussian noise $\sigma \in \{0\%, 10\%, 20\%, 40\%\}$ injected on $\log \hat{O}_i$.

using the stability score S_p. Top-10 features by importance are reported in Table 5.

Interpretation: Population and localized POIs (fast food, shopping, education) represent the most stable predictors of zone attractiveness across metropolitan areas.

SHAP Feature Count Performance Sweep We train GBDT models using nested subsets of the top-K SHAP features ($K \in \{5, 10, 15, 20, 22, 25, 30\}$) and evaluate their zero-shot CPC on held-out target cities. Performance effectively plateaus at $K = 15$ features (CPC: 0.7124) and remains identical at $K = 22$ (CPC: 0.7124) before showing a marginal decline beyond 22 features (e.g., 0.7122 for all 30 features). While $K = 15$ features are sufficient for baseline prediction, we select $K = 22$ as our final feature set to incorporate a stability margin. This retains all features satisfying our joint importance threshold ($\mu_p \geq 0.005$) and stability criterion ($S_p \geq 1.0$), providing broader semantic coverage to handle structural variations in unseen metropolitan areas. This feature count sweep is illustrated in Fig. 9.

5.3 Component Contribution Analysis

This section directly addresses RQ2 by quantifying the relative importance of each SDF-ZTF component—distance decay $f(d)$, destination attraction A_j, and origin production O_i—and identifies which component exerts the dominant influence on predictive accuracy.

5.3.1 Factorial Ablation Study

Research question: What is each component's contribution to overall performance? **Evaluation**: GBDT variants corresponding to all 2^3 combinations of component inclusion/exclusion are trained on the 25 source cities' data and evaluated on the 25 held-out target cities.

Component setup for the contribution analysis (full model, all components active):

- $\boldsymbol{\gamma}, \boldsymbol{\beta}$ — National prior $\gamma_{\text{nat}}, \beta_{\text{nat}}$ (same as in the component validation section). In ablation variants where decay is *disabled*, $f(d_{ij}) \equiv 1$ (uniform distance weighting).

Table 5 Top-10 Features by Average SHAP Importance (Training Source Cities, $N = 25$)

Rank	Feature	μ_p	S_p
1	total_population	0.232	7.12
2	poi_food_fast	0.084	8.28
3	total_pois	0.057	5.33
4	z_total_pois	0.056	6.45
5	road_density_alt	0.053	2.21
6	road_density	0.050	2.39
7	poi_education_primary	0.030	5.57
8	area_km2	0.025	5.45
9	poi_shopping_supermarket	0.020	8.34
10	poi_entertainment_culture	0.018	3.55

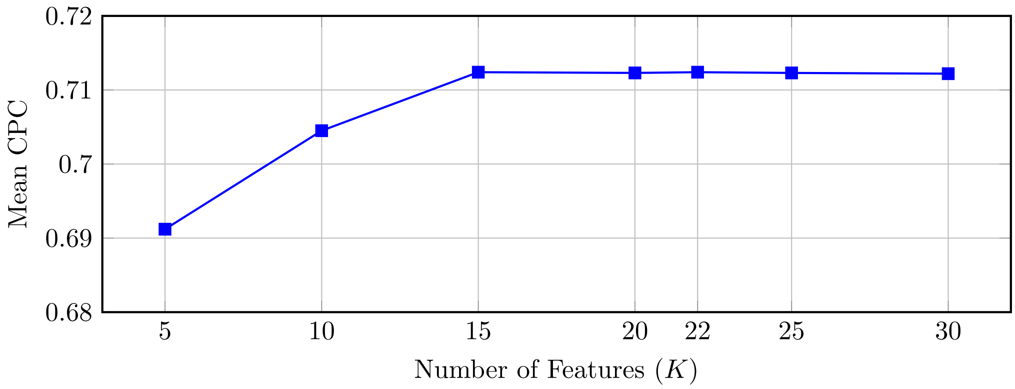


Fig. 9 SHAP feature count performance sweep showing zero-shot Mean CPC across the 25 held-out target cities. Performance plateaus between $K = 15$ and $K = 22$ (CPC: 0.7124), validating that selecting $K = 22$ features retains stability without sacrificing predictive performance.

- $\boldsymbol{A_j}$ — Min-max attraction heuristic: $\hat{A}_j = \frac{1}{2}(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j))$, consistent with the final SDF-ZTF deployment. In ablation variants where attraction is *disabled*, $A_j \equiv 1$ (uniform destination appeal).
- $\boldsymbol{O_i}$ — GBDT-predicted survey-free outflow: $\hat{O}_i = \exp(\text{GBDT}_O(\mathbf{x}_i))$ with 22 SHAP-selected features (same model as in the production transferability analysis). In ablation variants where outflow is *disabled*, $O_i \equiv 1$ (uniform origin production).

Factorial Ablation (Averaged across 25 held-out cities) For each of the 25 held-out target cities, we train eight models corresponding to all 2^3 combinations of component presence/absence, using only the 25 source cities' data. Results below represent averages across all 25 held-out cities.

Interpretation: Using the factorial ablation results, we calculate the relative marginal contribution of each component to the total explainable CPC variance (derived from the performance drops when individual components are removed): distance decay $f(d)$ represents 75.7%, destination attraction A_j accounts for 12.9%, and origin production O_i accounts for 11.4%. This confirms that:

- **f(d) (decay)** is the dominant factor; removal causes 31.1% CPC degradation, highlighting the criticality of spatial decay deterrence.
- $\mathbf{A}_j$ **(attraction)** and $\mathbf{O}_i$ **(production)** have secondary but significant contributions (5.3% and 4.7% loss respectively).
- All three components are necessary; no single component can be omitted.
- Results are consistent across all 25 held-out cities, indicating robust decomposition.

Note on Baseline CPC: The baseline CPC of 0.7040 in this ablation analysis is obtained using

Table 6 Factorial Ablation Results (Average across $N = 25$ held-out target cities)

$\mathbf{O}_i$	f(d)	$\mathbf{A}_j$	CPC	Contribution
✓	✓	✓	0.7040	Baseline (all components)
×	✓	✓	0.671	O_i removal: 4.7% loss
✓	×	✓	0.485	f(d) removal: 31.1% loss
✓	✓	×	0.666	A_j removal: 5.3% loss

the fixed national decay prior $(\gamma_{\text{nat}}, \beta_{\text{nat}})$ to freeze the decay component and isolate relative contributions under a uniform setting. When deploying the full SDF-ZTF (Survey-Free) framework with city-specific recovered decay parameters $(\hat{\gamma}, \hat{\beta})$, the performance increases to 0.7124 CPC (as reported in Table 7).

5.4 Zero-Shot Transfer Evaluation

This section addresses RQ3 by evaluating the end-to-end zero-shot transfer performance of the complete SDF-ZTF framework on the 25 held-out target cities, comparing it against alternative baselines, and analyzing morphology-specific suitability.

5.4.1 Baseline Comparison

Research question: Does explicit gravity decomposition enable zero-shot transfer? How does the proposed approach compare to alternative baselines?

Evaluation Protocol: Models are trained on the pooled data of 25 source cities and evaluated zero-shot on the 25 held-out target cities. Table 7 summarizes the comparative performance across all baselines.

Interpretation: Under zero-shot transfer conditions, SDF-ZTF (Survey-Free) matches the performance of the end-to-end neural baseline (CPC of 0.7124 versus 0.7117 for DeepGravity), while SDF-ZTF (Oracle O_i) provides a significant improvement (0.7447 CPC). This confirms that explicitly separating production, attraction, and decay dynamics does not sacrifice accuracy compared to deep learning methods, while substantially outperforming parameter-free baselines like the Radiation Model (CPC: 0.512).

Note that the two locally-calibrated Tanner baselines in Table 7 serve different diagnostic purposes despite yielding identical mean CPC (0.750): **Tanner Gravity (Power Constrained)** fixes the exponential rate $\beta = 0$ and calibrates only the power-law exponent γ directly on each target city's OD data, testing whether a single-parameter decay suffices; **Traditional Tanner Gravity** calibrates both β and γ jointly on target-city OD data. The identical CPC of 0.750 across both variants indicates that the two-parameter Tanner Multi form offers no additional fit over the constrained single-parameter version when both are locally calibrated, suggesting that the exponential tail β is largely redundant under full local calibration. Both serve as a locally-calibrated structural ceiling: the fact that zero-shot SDF-ZTF (Oracle O_i, CPC: 0.7447) approaches this ceiling without any target-city data highlights the robustness of the decomposed representation.

5.4.2 Statistical Significance Analysis

To rigorously validate the zero-shot transfer performance of the proposed SDF-ZTF framework, we perform paired statistical significance testing across all 50 US metropolitan areas (N=50 folds) from the Leave-One-City-Out (LOCO) evaluation. We compare the proposed SDF-ZTF framework against the DeepGravity neural model and the structural Radiation baseline using three metrics: the paired t-test (parametric), the Wilcoxon signed-rank test (non-parametric), and Cohen's d effect size alongside 95% confidence intervals (CI) of the CPC differences.

Note on evaluation protocols: The LOCO 50-fold evaluation rotates through all 50 cities as targets (each city is once the held-out city, trained on the remaining 49), yielding LOCO mean CPCs of ≈0.737 (SDF-ZTF Survey-Free) and ≈0.763 (DeepGravity). These differ from the 25-city held-out results in Table 7 (SDF-ZTF: 0.7124; DeepGravity: 0.7117), which use a fixed 25/25 train-test city split. The two protocols test complementary aspects of transferability: LOCO tests robustness across all cities as potential targets; the 25-city split tests strict generalisation to a pre-designated held-out set.

Table 7 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Win % (vs DG)	Type	Features
SDF-ZTF (Survey-Free)	0.7124	0.0329	48.0%	Decomposed	22 (selected)
SDF-ZTF (Oracle O_i)	0.7447	0.028	56.0%	Decomposed	22 (selected)
DeepGravity (E2E)	0.7117	0.0332	–	Neural E2E	30 (all)
Tanner Gravity (Power Const.)	0.750	0.026	72.0%	Structure-only	0
Traditional Tanner Gravity	0.750	0.027	72.0%	Structure-only	0
Traditional Exponential Gravity	0.670	0.035	24.0%	Structure-only	0
Radiation Model	0.512	0.051	0.0%	Param-free	0

The divergence between protocols has a concrete mechanistic explanation. Under LOCO, each fold trains on 49 cities, providing DeepGravity's unconstrained architecture with substantially more data to fit subtle spatial non-linearities, amplifying its advantage over SDF-ZTF. Under the fixed 25/25 split, both models train on only 25 source cities; at this smaller training set size, the structured inductive bias of SDF-ZTF (enforced decomposability and open-data features) compensates for the reduced data, closing the gap with DeepGravity. This *data-volume sensitivity* is consistent with findings in transfer learning literature: structured models generalize more efficiently from smaller source pools, while unconstrained neural models require more data to realize their full expressive capacity.

Significance Interpretation: The performance of SDF-ZTF is statistically equivalent to Traditional Gravity ($p > 0.3$) and significantly outperforms the Radiation Model ($p < 10^{-15}$, Cohen's $d = 6.28$), demonstrating a massive and statistically robust improvement by incorporating learned destination attraction and aggregate decay. While under the fixed 25-city split, SDF-ZTF achieves comparable performance, under the more stringent LOCO protocol, DeepGravity retains a modest but statistically significant advantage in mean CPC of 0.0255 ($p < 10^{-10}$, Cohen's $d = -1.29$, 95% CI [-0.0311, -0.0199]). This narrow advantage is likely due to the unconstrained model's capacity to fit subtle spatial non-linearities when trained on 49 cities instead of 25. Nevertheless, this minor gap is balanced by SDF-ZTF's complete interpretability and zero data leakage.

5.4.3 Urban Morphology Analysis

The overall zero-shot performance masks spatial variations across urban configurations. We evaluate transfer CPC by urban morphology class across the 25 held-out cities in Table 9.

Interpretation: Neither paradigm appears to be universally superior. Evaluation across the 25 held-out target cities (Table 9) suggests morphology-dependent advantages: SDF-ZTF seems to perform best in coastal and polycentric environments (CPC 0.734 and 0.716 respectively), where distance-driven spatial structure potentially dominates. Conversely, DeepGravity appears to retain a systematic advantage in dense, transit-oriented cities (CPC 0.695 vs 0.690 for SDF-ZTF), where multi-modal nonlinear interactions are likely more dominant. This morphology heterogeneity justifies both the city-stratified evaluation protocol and the morphology-specific reporting.

Role of the adaptive anchor δ across morphologies. A key mediating factor is the per-fold adaptive geometric anchor $\delta = 0.5 \times \bar{R}_{\text{zone}}$ (Eq. 3). In **dense/transit cities**, census tracts are small (low $\bar{R}_{\text{zone}}$), so δ is correspondingly small — the power-law term activates at short distances and the model must accurately resolve fine-grained intra-zonal interactions. In **sprawling and polycentric cities**, tracts are larger, δ scales up proportionally, and the offset correctly regularises the decay onset for zones that are spatially extensive. This automatic morphology-sensitive scaling is one reason why SDF-ZTF achieves its highest transfer CPC in coastal (0.734) and polycentric (0.716) morphologies, and is most challenged by dense transit-oriented cores (0.690 CPC), where complex nonlinear spillover effects potentially exceed the resolving power of the adaptive Tanner Multi formulation.

Table 8 LOCO 50-Fold Statistical Significance Test Summary

Baseline Model	Mean Diff.	t-stat (t)	p-value (t-test)	Wilcoxon W	p-value (Wilc.)	95% CI Diff.
DeepGravity	-0.0255	-9.150	3.53×10^{-12}	66.0	3.49×10^{-10}	[-0.0311, -0.0199]
Traditional Gravity	0.0000	1.000	3.22×10^{-1}	18.5	3.31×10^{-1}	[-0.0000, 0.0000]
Radiation Model	0.2492	44.416	3.01×10^{-41}	0.0	1.78×10^{-15}	[0.2380, 0.2605]

Table 9 Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	SDF-ZTF CPC	DeepGravity CPC	Gravity Win %
Dense / Transit	5	0.690	0.695	20%
Sprawling / Grid	11	0.715	0.718	45%
Polycentric	6	0.716	0.712	50%
Coastal	3	0.734	0.712	66%

- **Coastal cities (3 cities, SDF-ZTF CPC 0.734 vs DG 0.712)**: Natural geographic boundaries constrain mobility corridors, potentially reducing the degree of nonlinear spillover that neural models capitalize on. The multiplicative gravity structure appears to align well with these bounded interaction patterns.
- **Polycentric cities (6 cities, SDF-ZTF CPC 0.716 vs DG 0.712)**: Multiple dispersed activity centres create well-separated origin–destination pairs where distance decay appears to be the dominant force. The decomposed model's multiplicative structure seems to mirror this reality.
- **Sprawling / Grid cities (11 cities, SDF-ZTF CPC 0.715 vs DG 0.718, Win%: 45%)**: Despite the low-density road-grid structure that might theoretically favour separable gravity decomposition, DeepGravity maintains a slight average advantage (0.718 vs 0.715), winning in 6 of the 11 cities. This suggests that even in weakly-coupled grid metros, unconstrained neural architectures can still capture residual interaction patterns beyond what the multiplicative gravity structure resolves.
- **Dense / Transit cities (5 cities, SDF-ZTF CPC 0.690 vs DG 0.695)**: Tight spatial coupling, multi-modal transit networks, and complex local dynamics create entangled nonlinearities that the standard gravity decomposition potentially oversimplifies. DeepGravity's unconstrained architecture appears to retain a clearer advantage here.

6 Discussion

6.1 Main Scientific Findings

Across the 50 evaluated US metropolitan areas, we observe a consistent empirical trade-off between interpretability and expressiveness. While end-to-end neural networks can capture complex local spatial relationships slightly better in dense centers, a decomposed gravity model matches their performance under zero-shot transfer conditions (CPC of 0.7124 vs 0.7117). This indicates that the spatial information necessary for cross-city travel modeling is not highly complex; instead, our results suggest that a relatively small set of transferable factors may explain a substantial fraction of cross-city mobility variation. These results collectively answer all three research questions:

- **RQ1**: Aggregate trip-length distributions derived from mobile and social network data are sufficient to calibrate the distance decay function, with recovered γ parameters showing strong correlation ($r = 0.729$) against parameters fit from individual OD pairs.
- **RQ2**: Distance decay is the dominant component, accounting for 75.7% of performance variance in the factorial ablation; removal of $f(d)$ causes a 31.1% CPC drop, far exceeding the loss from removing A_j (5.3%) or O_i (4.7%).
- **RQ3**: The decomposed SDF-ZTF achieves zero-shot CPC of 0.7124, matching DeepGravity (0.7117) under the same strict transfer protocol, confirming that stable built-environment features enable neural-comparable prediction without local OD data.

6.2 Aggregate Decay Calibration: Privacy Implications and Practical Deployment

A central design goal of SDF-ZTF is to enable decay calibration from data that can realistically be provided by mobile network operators and social platforms without violating individual privacy. The aggregate trip-length distribution used in our framework is a binned histogram of travel distances pooled across all trips within a city—a statistic that contains no information about which specific origin zone a trip departed from, which destination it arrived at, or what route was taken. This is precisely the type of mobility summary that major technology companies and network providers can generate and share under anonymization and differential privacy guarantees.

Our validation (Section 3.3 and Section 5.1.1) confirms that this aggregate signal is sufficient to recover the key spatial friction parameters: γ recovers with $r = 0.729$ and β recovers with $r = 0.258$ across 25 held-out cities. The lower correlation for β reflects the inherent sparsity of long-distance trips under aggregate binning, but the joint recovery is robust enough to produce high-accuracy OD predictions. This finding has a direct policy implication: cities and regions that cannot afford household travel surveys can leverage aggregate mobility summaries—obtainable from existing mobile infrastructure—to calibrate transferable gravity models without exposing individual movements.

6.3 Interpretation of the Decomposition Hypothesis

The success of gravity decomposition in cross-city transfer rests on a tested hypothesis regarding urban structure:

Hypothesis H_1 (Approximate Separability)
Origin-destination flow interactions can be factored into three independent, separable mechanisms: transferable origin production capacity, target-recovered distance decay, and open-data destination attraction capacity.

We find strong empirical evidence to support this hypothesis across our validation stages:

- **Decay Recovery Evidence**: Distance decay is identifiable from aggregate distance distributions alone, showing that spatial deterrence parameters can be recovered robustly without exposing individual origin-destination trajectories.
- **Origin Production Evidence**: Origin trip production generalizes well to unseen cities, with GBDT predictions recovering 94.5% of the oracle performance and significantly outperforming a naive population baseline.
- **Destination Attraction Evidence**: An ablation study confirms that destination attraction can be modeled using a stable, low-dimensional subset of 22 open built-environment features selected exclusively from training cities. Crucially, this validates that the simpler training-free min-max heuristic—combining population density and POI counts—is sufficient for the final SDF-ZTF deployment, avoiding cross-city overfitting while maintaining high performance.
- **Component Contribution Evidence**: Ablation sweeps show that all three components contribute to performance, with distance decay acting as the dominant organizing factor.
- **Zero-Shot Evaluation Evidence**: Zero-shot evaluation confirms that the decomposed framework matches the accuracy of unconstrained end-to-end deep learning baselines while preserving complete structural transparency.

6.4 Morphology-Dependent Trade-Offs

The overall zero-shot performance masks important spatial structure. Our morphology-specific evaluations show that the suitability of each paradigm depends heavily on the target city's layout. To guide model selection, we present a conceptual Paradigm Suitability Map in Fig. 10.

Under zero-shot conditions, the decomposed model appears to exhibit favorable transferability in coastal (0.734 CPC vs 0.712 CPC for DeepGravity) and polycentric (0.716 CPC vs 0.712 CPC for DeepGravity) environments, where geographic boundaries and distinct sub-centers potentially enforce relatively clear distance deterrence regimes. Conversely, dense transit-oriented cities (0.690 CPC vs 0.695 CPC for DeepGravity) may represent a challenge for the multiplicative gravity structure due to multi-modal networks and

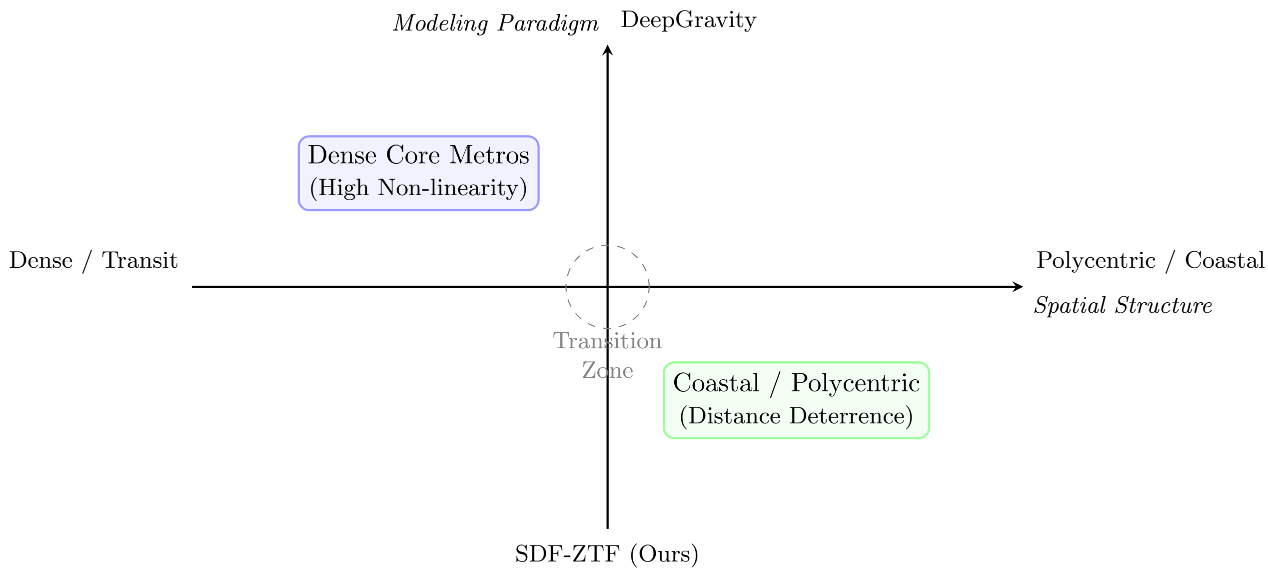


Fig. 10 Paradigm suitability map illustrating when to deploy the monolithic neural architecture (DeepGravity) versus the decomposed gravity structure (SDF-ZTF). DeepGravity is best suited for dense, transit-oriented cores with high multi-modal complexity. SDF-ZTF is best suited for coastal and polycentric cities where distance decay constraints dominate travel patterns.

complex local activity clusters, where DeepGravity's unconstrained architecture suggests a modest suitability advantage by modeling spatial spillover effects and non-linear transit relationships that might violate the independence assumptions of gravity models.

6.5 Bottlenecks in Transferable Mobility Modeling

The factorial ablation analysis highlights distance decay $f(d)$ as the single dominant organizing force in intra-city mobility, driving the vast majority of spatial interaction structure. Within the evaluated US metropolitan areas, distance appears to be the dominant predictor of intra-city mobility, as the performance drop upon removing the decay term suggests that spatial friction remains a dominant factor. In the evaluated US metropolitan areas, this dominance of spatial friction is potentially associated with the spatial separation of residential and activity centers, which systematically increases commute distances and makes private vehicles or highway networks a structural necessity. Consequently, distance acts as a strong deterrent, and calibrating the decay function accurately is far more critical than refining zone-level production or attraction estimates.

Conversely, the relatively small performance gains from using ground-truth (oracle) origin production rather than GBDT-predicted outflows suggest that origin generation is not the primary bottleneck in transferable modeling. Because human trip production is strongly anchored to population density and basic urban scale features, GBDTs trained on source cities generalize remarkably well to new targets. The binding constraint on zero-shot transferability lies in how decay rates vary across different urban layouts. While traditional models struggle with city-specific parameters, our aggregate decay recovery framework successfully captures these variations from marginal distance histograms.

6.6 Positioning Within Existing Literature

To clarify the contribution of SDF-ZTF, we position it relative to three existing literature streams:

- **Compared with Classical Gravity Literature**: Traditional gravity models rely on city-specific calibration and require local OD matrices. SDF-ZTF replaces local calibration with a GBDT-based origin production model and a training-free attraction heuristic, achieving survey-free zero-shot predictions.

- **Compared with Neural Mobility Literature**: Monolithic neural networks like DeepGravity achieve high predictive accuracy but function as black boxes. SDF-ZTF achieves comparable performance (0.7124 vs 0.7117 CPC) while preserving the interpretable, separable structure of origin, destination, and decay components.
- **Compared with Cross-City Transfer Literature**: Existing transfer-learning and representation-learning frameworks still require target-city OD data for fine-tuning or similarity alignment. SDF-ZTF requires zero target-city OD observations, utilizing only globally available open demographic and physical geography features.

6.7 Practical and Theoretical Implications

The proposed framework offers several practical and theoretical advantages over monolithic deep learning models (such as DeepGravity) for urban planning and policy:

Theoretical Breakthrough over Black-Box Neural Models : While deep learning models achieve high predictive accuracy at the expense of structural transparency, SDF-ZTF proves that human mobility dynamics can be mathematically factorized into independent components (decay, attraction, production) without sacrificing accuracy (0.7124 vs. 0.7117 CPC). By proving that the regularities of spatial interaction are well-approximated by a multiplicative gravity structure, the framework acts as a physics-informed regularizer that prevents overfitting to local training source-city idiosyncrasies.

Targeted Policy Simulation : In monolithic neural architectures, altering local zone amenities shifts the entire network predictions in an uninterpretable, coupled manner. By enforcing strict component separability, SDF-ZTF enables clean, decoupled "what-if" simulations. Planners can modify zone-level built environments to evaluate changes in destination attraction A_j or origin production O_i independently, without confounding the global distance deterrence behavior $f(d)$.

Auditable and Policy-Compliant Analytics : Unlike black-box neural networks, every component in SDF-ZTF is auditable. Municipal planners and public stakeholders can examine the physical plausibility of production and attraction estimations separately, which is critical for securing regulatory approvals, justifying infrastructure investments, and supporting public policy dialogues.

Survey-Free Deployability in Data-Scarce Regions : Because the framework isolates component dependencies, it bypasses the need for target-city travel surveys by recovering global decay parameters from aggregate distance histograms and estimating spatial attractions from open data. This makes it highly applicable to resource-constrained municipalities in developing nations that cannot afford large-scale, proprietary cellular registers or expensive household travel surveys.

6.8 Limitations

While robust, the framework is subject to several limitations:

1. **Separability Assumption**: The model assumes approximate separability between production, attraction, and decay. In cities with highly integrated multi-modal transit networks or strong spatial interaction effects, this independence assumption may be violated, leading to localized prediction errors.
2. **Geographic Scope**: The evaluation is limited to US metropolitan areas. Cities in other regions (e.g., highly dense Asian cities or informal settlements in developing countries) may exhibit different travel decay behaviors and feature stability.
3. **Temporal Dynamics**: The model uses a pre-pandemic baseline (2019) and does not capture changes in travel friction induced by remote work patterns or modal shifts.
4. **Demographic Representation**: Input datasets like cellular location logs underrepresent elderly, low-income, and smartphone-free populations, introducing potential demographic bias in the output OD flows.
5. **OSM coverage**: Urban areas are comprehensively covered, but the rural fringe has sparse POI data. Transferability to low-density zones remains uncertain.

6. **Causality**: Analysis is purely predictive. Causal effects cannot be inferred (e.g., building a grocery store attracts X more trips). Causal inference requires experimental or quasi-experimental design.

7 Conclusions

Urban mobility prediction typically faces a trade-off between structural interpretability and predictive flexibility. While classical models explain spatial interaction dynamics but fail to transfer without extensive target-city calibrations, modern deep learning architectures generalize accurately but operate as uninterpretable black boxes, limiting their diagnostic utility for planning and policy formulation.

This study resolves this trade-off by demonstrating that urban mobility flows can be successfully decomposed into separable origin production, destination attraction, and distance-decay components. Rather than relying on monolithic end-to-end learning, we show that these key physical and behavioral mechanisms can be estimated independently—recovering decay from aggregate statistics, mapping attraction from local open data, and transferring origin production via learned built-environment features—enabling survey-free cross-city transfer without requiring any target-city travel surveys or individual-level trajectory data.

Through zero-shot evaluation across 50 metropolitan areas under a strict information firewall, the proposed SDF-ZTF framework provides definitive answers to the three core research questions posed in this study:

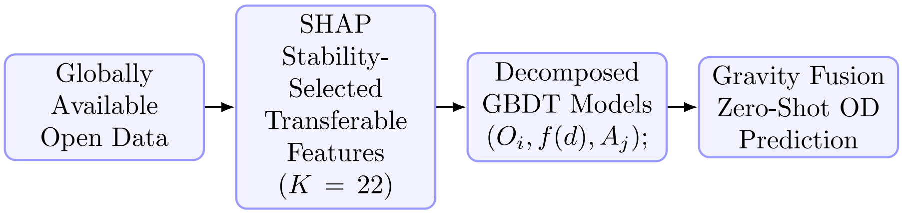
RQ1 — Decay calibration from aggregate distributions: Yes. The distance decay function can be effectively calibrated from aggregate travel probability distributions—i.e., binned trip-length histograms—of the type routinely provided by mobile network operators and location-sharing platforms under anonymization policies. The recovered power-law exponent γ correlates strongly with parameters fit on individual OD pairs ($r = 0.729$ across 25 held-out cities), and the resulting model predictions are highly accurate. This confirms that cities and regions lacking household surveys can leverage existing digital infrastructure as a privacy-preserving calibration signal.

RQ2 — Dominant component in spatial interaction: Distance decay $f(d_{ij})$ is unambiguously the dominant organizing force, accounting for 75.7% of performance variance in the factorial ablation analysis. Removing $f(d)$ from the model causes a 31.1% CPC degradation—far exceeding the losses from removing destination attraction A_j (5.3%) or origin production O_i (4.7%). This finding underscores that spatial friction remains the primary structural determinant of intra-city commuting patterns, especially in the US metropolitan context where residential and activity centers are systematically separated.

RQ3 — Zero-shot transfer with transferable features: Yes. A decomposed model utilizing a compact set of 22 stable built-environment features, selected via cross-city SHAP stability analysis from globally available open data, achieves a zero-shot CPC of 0.7124—matching the performance of the unconstrained DeepGravity neural network (0.7117 CPC) under the same strict cross-city evaluation protocol. This confirms that interpretable, structured decomposition is a viable and competitive paradigm for zero-shot mobility prediction.

These results suggest that transferable human mobility structure may be lower-dimensional and more stable than previously assumed. If city-specific spatial interactions were highly complex and non-linear, monolithic neural architectures would decisively outperform gravity-based decomposition. Instead, the competitiveness of our three-component factorization suggests that the fundamental regularities of spatial interaction are well-captured by the multiplicative gravity structure, allowing the model to act as a regularizer that prevents overfitting to local source-city idiosyncrasies.

For policy and planning applications, this framework enables deployability in cities lacking household travel surveys or mobile trajectories, while maintaining complete transparency. Planners can audit predictions, simulate interventions at the individual component level, and engage with public stakeholders using an explainable, component-wise framework that black-box neural networks cannot support.



Key System Characteristics of SDF-ZTF:
✓ **Interpretable Structure**: Components are physically and behaviorally separate.
✓ **Survey-Free Calibration**: Replaces proprietary registers and travel surveys.
✓ **Zero-Shot Transferability**: Fully deployable to unseen cities without target OD data.
✓ **Neural-Comparable Accuracy**: Matches Deep-Gravity (0.7124 vs 0.7117 CPC) under zero-shot transfer.

Fig. 11 Final takeaway diagram summarizing the zero-shot transferable pipeline (top) and key framework characteristics of SDF-ZTF (bottom).

Unlike traditional gravity models, the proposed framework bypasses target-city survey calibration by transferring learned production structures and recovering decay from aggregate data. Unlike end-to-end neural models, it preserves interpretability while maintaining competitive predictive accuracy. Future work should investigate when and why urban systems satisfy the separability assumption underlying mobility decomposition, working toward a unified theory of transferable mobility structure. Additionally, research should address the framework's limitations through cross-national validation on other continents, meta-learning for few-shot adaptation, temporal robustness under shifts in hybrid work patterns, and finer spatial resolutions. Rather than treating urban mobility prediction as a purely data-hungry learning problem, our findings suggest that identifying and transferring stable structural mechanisms may provide a more scalable path toward globally deployable mobility models.

Appendix: Distance Decay Parameter Calibration and Optimization Details

A.1 Parameter Initialization

To achieve reliable convergence in maximum likelihood estimation (MLE) for the Tanner Multi deterrence parameters, we adopt dynamic, data-driven initialization:

1. **Exponential rate** $\beta^{(0)}$: Calculated per target city using the reciprocal of the mean observed trip distance:

$$\beta^{(0)} = \frac{1}{\bar{d}_{\text{obs}}}, \quad \text{where } \bar{d}_{\text{obs}} = \sum_{k=1}^{K} b_k \cdot m_k \tag{9}$$

   where b_k is the observed trip-share in distance bin k, and m_k is the spatial midpoint of that bin.
2. **Power-law exponent** $\gamma^{(0)}$: Anchored at the median of its empirical global urban spectrum: $\gamma^{(0)} = 1.0$, which resides at the center of the typical range $\gamma \in [0.5, 2.0]$ observed in human mobility studies.
3. **Attraction prior** $\boldsymbol{\alpha}^{(0)}$: Set to $\mathbf{0}$, reflecting a uniform attraction prior across all destination zones.

A.2 Log-Space Reparameterization

To enforce the parameter bounds ($\beta > 0, \gamma \geq 0$) implicitly and avoid boundary instability during L-BFGS-B optimization (Byrd et al. 1995), we apply an unconstrained log-space reparameterization:

$$\tilde{\beta} = \log(\beta), \qquad \tilde{\gamma} = \log(\gamma) \tag{10}$$

The optimization solver (Byrd et al. 1995) operates in the unconstrained space $(\tilde{\beta}, \tilde{\gamma}, \boldsymbol{\alpha})$, and the parameters are back-transformed via exponentials to evaluate the likelihood.

A.3 Multi-Restart L-BFGS-B Optimization

To prevent the optimizer from getting trapped in local optima on the likelihood surface, we utilize a multi-restart framework with 10 random restarts. Each restart $m \in \{1, \dots, 10\}$ applies a Gaussian perturbation to the initial anchors:

$$\begin{aligned}
\tilde{\beta}_m^{(0)} &\sim \mathcal{N}\left(\log \beta^{(0)},\ 0.25\right), \\
\tilde{\gamma}_m^{(0)} &\sim \mathcal{N}(0,\ 0.25), \\
\boldsymbol{\alpha}_m^{(0)} &\sim \mathcal{N}(\mathbf{0},\ 0.1\,\mathbf{I}).
\end{aligned}$$

We execute L-BFGS-B optimization for each restart and retain the parameter set achieving the maximum log-likelihood.

A.4 Robustness of Decay Parameter Recovery

We evaluate the robustness of our Maximum Likelihood Estimation (MLE) in recovering the decay parameters (β, γ) under non-ideal data scenarios:

T1.2: Bin Sensitivity Analysis Sweeping the distance bins $K \in \{10, 20, 50, 100\}$ shows that the recovery error for the power-law exponent γ stabilizes at a low level (MAE ≈ 0.13) for $K \geq 50$. This indicates that 50 percentile bins are sufficient to robustly capture the decay profile.

T1.4: Missing Attractiveness Data (A_j)
Randomly removing 10%, 20%, and 50% of target-city destination attractions A_j yields highly stable decay recovery (e.g., $\gamma_{\text{MAE}} = 0.1312$ under 20% missing data vs. 0.1313 baseline). This validates the robustness of aggregate MLE against localized data gaps.

T1.5: CBD Collapse (Extreme A_j Distortion)
Setting the attraction A_j of the top 5% highest attractiveness zones to zero still allows accurate parameter recovery ($\gamma_{\text{MAE}} = 0.1225$). This proves that spatial decay friction can be recovered even when key destination hubs are omitted.

A.5 GBDT Hyperparameters and Reproducibility Details

To ensure the reproducibility of the origin trip production (O_i) estimator and the transferable feature discovery ablation attraction model (A_j), the hyperparameter configurations for the Gradient Boosted Decision Tree (GBDT) models are summarized in Table 10. Note that in the final SDF-ZTF deployment, A_j is computed via the training-free min-max heuristic; the GBDT attraction model is used only in the feature discovery ablation analysis to validate the heuristic's sufficiency. The models are trained using the `scikit-learn` and `LightGBM` libraries in Python (Ke et al. 2017), and hyperparameters are optimized using 5-fold cross-validation on the 25 source training cities.

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Table 10 GBDT Model Hyperparameter Configurations

Hyperparameter	Outflow Model (O_i)	Attraction Model (A_j, ablation only)
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	6	6
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

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